Full proofs

Online Hypervolume Subset Selection under Semi-Bandit Feedback

This document reproduces Appendix A of the manuscript and provides the full proofs of Theorems 1–3. Numbering is identical to that of the manuscript: sections, lemmas and equations prefixed by "A" are in this document, while the other section, algorithm and equation numbers refer to the main text of the manuscript.

A. Upper-bound proofs

This appendix establishes the sublinear α-regret guarantee for *THV-UCB* announced in Section 5. We prove two complementary statements:

- a *gap-free* bound (Theorem 1), of order $\tilde{O}(d\sqrt{nkT})$, valid on *every* instance regardless of how close the arms are to each other;
- a *gap-dependent* bound (Theorem 2), of order $\tilde{O}(d^2nk^{2.5}/\Delta_{\min})$, which becomes polylogarithmic in T as soon as the instance is well separated.

Both proofs share the same first steps (Appendices A.1 to A.5), which reduce $\bar{R}_T$ to a sum of confidence radii, in (A.3). They diverge only in *how* this sum is controlled: the gap-free bound (Appendix A.6) treats every pull anonymously via Cauchy–Schwarz over the whole horizon; the gap-dependent bound (Appendix A.7) opens up the greedy construction stage by stage, tracks the actual comparisons $\Delta_t^{\mathrm{UCB}}(\cdot \mid S)$ performed by the algorithm, and applies the same Cauchy–Schwarz idea *locally*, to each stage's deviation rounds. We discuss what each proof technique does, and does not, use about *THV-UCB* in Appendix A.9.

A.1. Preliminaries

Lemma A.1 (Monotonicity and submodularity of hypervolume)**.** *For any fixed reference point* $\mathbf{r}$ *dominated by all achievable means, the set function* $S \mapsto \mathrm{HV}(S)$ *is monotone (non-decreasing) and submodular.*

Proof. Recall $\mathrm{HV}(S) = \lambda_d\big(\bigcup_{i\in S}\prod_{j=1}^{d}[r_j, \mu_{i,j}]\big)$.

Monotonicity. If $S \subseteq T$, the union of boxes only grows, so $\mathrm{HV}(S) \le \mathrm{HV}(T)$.

Submodularity. Let $A \subseteq B$ and $i \notin B$. Write $V_A = \bigcup_{j\in A}\prod_{\ell}[r_\ell, \mu_{j,\ell}]$ and $V_i = \prod_{\ell}[r_\ell, \mu_{i,\ell}]$. Then

$$\mathrm{HV}(A \cup \{i\}) - \mathrm{HV}(A) = \lambda_d(V_i \setminus V_A), \qquad \mathrm{HV}(B \cup \{i\}) - \mathrm{HV}(B) = \lambda_d(V_i \setminus V_B).$$

Since $V_A \subseteq V_B$, $V_i \setminus V_B \subseteq V_i \setminus V_A$, hence submodularity. □

Remark. *The same argument, applied verbatim with* $u_{i,j}(t)$ *in place of* $\mu_{i,j}$ *(the proof never uses anything about* $\mu_{i,j}$ *beyond it being a real number), shows that for every fixed* t, $S \mapsto \mathrm{HV}_t^{\mathrm{UCB}}(S)$ *(defined in Appendix A.4) is also monotone and submodular.*

Lemma A.2 (Coordinate-wise concentration)**.** *Assume each coordinate is* η*-sub-Gaussian. Then with probability at least* $1-\delta$*, for all* i, j, t*,*

$$\left|\widehat{\mu}_{i,j}(t) - \mu_{i,j}\right| \le \beta_i(t) = \sqrt{\frac{2\eta\log(ndt^2/\delta)}{\max\{1, N_i(t-1)\}}}.$$

Proof. Fix i, j and a pull count $s = N_i(t-1) \ge 1$. By the sub-Gaussian Hoeffding bound, $\mathbb{P}\big(|\widehat{\mu}_{i,j}(t) - \mu_{i,j}| > \sqrt{2\eta\log(x)/s}\big) \le 2/x$. Taking $x = ndt^2/\delta$ and a union bound over $i \in [n]$, $j \in [d]$, and $t \in [T]$ (using $\sum_{t\ge1} t^{-2} \le 2$) gives total failure probability $O(\delta)$; absorbing the constant into δ yields the stated bound. □

We write $\mathcal{E}$ for the event on which Lemma A.2 holds, so $\mathbb{P}(\mathcal{E}) \ge 1-\delta$. After the initialization phase of Algorithm 1 (which lasts at most $t_0 = \lceil nm_0/k\rceil$ rounds), every arm has $N_i(t-1) \ge m_0 \ge 1$, so the $\max\{1, \cdot\}$ in $\beta_i(t)$ is never active for $t > t_0$; we drop it below for readability.

A.2. A shared greedy-approximation theorem

The following classical fact [1] is invoked twice in this appendix, for two different monotone submodular functions; we state it once to avoid duplicating the argument and, crucially, to keep visually distinct the two objects it produces.

Fact A.1 (Greedy approximation for monotone submodular maximization)**.** *Let $f : 2^{[n]} \to \mathbb{R}$ be monotone and submodular, and let $G = \{g_1, \dots, g_k\}$ be built by greedy maximization of marginal f-gain under the cardinality constraint k (i.e. $g_\ell \in \arg\max_{i \notin G_{\ell-1}} f(G_{\ell-1} \cup \{i\}) - f(G_{\ell-1})$, $G_\ell = G_{\ell-1} \cup \{g_\ell\}$, $G_0 = \emptyset$, $G = G_k$). Then*

$$f(G) \;\geq\; \left(1 - \frac{1}{e}\right) \max_{|S|=k} f(S) \;=\; \alpha \max_{|S|=k} f(S).$$

We will apply Fact A.1 to two different functions:

- $f = \mathrm{HV}_t^{\mathrm{UCB}}$ (monotone submodular by the previous remark), with $G = S_t$ the set actually built by *THV-UCB* at round t — this yields Corollary A.1 below, used in the reduction step;
- $f = \mathrm{HV}$ **directly on the true means** — a purely deterministic statement requiring no concentration event — with $G = G^\star$, a benchmark sequence introduced in Appendix A.7 and **distinct from the regret's true optimum** $S^\star$.

Corollary A.1 (Greedy approximation under optimism)**.** *For every round t, $\mathrm{HV}_t^{\mathrm{UCB}}(S_t) \geq \alpha\, \mathrm{HV}_t^{\mathrm{UCB}}(S_t^{\mathrm{opt}})$, where $S_t^{\mathrm{opt}} \in \arg\max_{|S|=k} \mathrm{HV}_t^{\mathrm{UCB}}(S)$.*

A.3. Safe pruning is lossless

Recall that, at every round t of its main loop, Algorithm 1 computes the lower vector $\mathbf{L}_i(t) = \widehat{\boldsymbol{\mu}}_i(t-1) - \beta_i(t)\mathbf{1}_d$ and the clipped optimistic vector $\mathbf{u}_i(t)$ of every arm, discards every arm i for which some $j \neq i$ satisfies $\mathbf{u}_i(t) \preceq \mathbf{L}_j(t)$, and runs the greedy step on the remaining candidate set A_t (with $A_t = [n]$ whenever fewer than k arms remain). The optimistic hypervolume $\mathrm{HV}_t^{\mathrm{UCB}}$ is the one defined in Section 4. We assume $n > k$.

Lemma A.3 (Safe pruning)**.** *For every round t of the main loop of Algorithm 1:*

(i) (deterministic) for every $S \subseteq [n]$ with $|S| = k$, there exists $S' \subseteq A_t$ with $|S'| = k$ and $\mathrm{HV}_t^{\mathrm{UCB}}(S') \geq \mathrm{HV}_t^{\mathrm{UCB}}(S)$; consequently,

$$\max_{S \subseteq A_t,\, |S|=k} \mathrm{HV}_t^{\mathrm{UCB}}(S) \;=\; \max_{S \subseteq [n],\, |S|=k} \mathrm{HV}_t^{\mathrm{UCB}}(S);$$

(ii) on $\mathcal{E}$, if the greedy maximizer is unique at every stage of the construction of $G^\star$, then $G^\star \subseteq A_t$.

Proof. If $A_t = [n]$ there is nothing to prove, so we assume that A_t is the set of arms that are not discarded, with $|A_t| \geq k$.

(i) Write $i \rhd j$ when $j \neq i$ and $\mathbf{u}_i(t) \preceq \mathbf{L}_j(t)$, so that i is discarded if and only if $i \rhd j$ for some j. Since rewards lie in $[0,1]^d$ and $\beta_j(t) > 0$, every coordinate of $\mathbf{L}_j(t)$ is strictly smaller than the corresponding coordinate of $\mathbf{u}_j(t)$. Hence $i \rhd j$ and $j \rhd l$ give $\mathbf{u}_i(t) \preceq \mathbf{L}_j(t) \preceq \mathbf{u}_j(t) \preceq \mathbf{L}_l(t)$, and $l \neq i$ because $\mathbf{u}_i(t) \preceq \mathbf{L}_i(t)$ is impossible: $\rhd$ is transitive and irreflexive, i.e. a strict partial order on the finite set $[n]$. Every discarded arm i therefore lies below a $\rhd$-maximal arm $\varphi(i)$, which is not discarded, i.e. $\varphi(i) \in A_t$, and which satisfies $\mathbf{u}_i(t) \preceq \mathbf{L}_{\varphi(i)}(t) \preceq \mathbf{u}_{\varphi(i)}(t)$. Set $\varphi(i) = i$ for $i \in A_t$. Given S with $|S| = k$, the set $\varphi(S) \subseteq A_t$ has at most k elements and, since the optimistic box of each $i \in S$ is contained in that of $\varphi(i)$, $\mathrm{HV}_t^{\mathrm{UCB}}(\varphi(S)) \geq \mathrm{HV}_t^{\mathrm{UCB}}(S)$. Completing $\varphi(S)$ with arbitrary arms of A_t up to size k (possible since $|A_t| \geq k$) yields S'; by monotonicity of $\mathrm{HV}_t^{\mathrm{UCB}}$, its optimistic hypervolume is at least that of $\varphi(S)$.

(ii) Suppose that some $i = i_\ell^\star \in G^\star$ is discarded, i.e. $\mathbf{u}_i(t) \preceq \mathbf{L}_j(t)$ for some $j \neq i$. On $\mathcal{E}$, $\boldsymbol{\mu}_i \preceq \mathbf{u}_i(t)$ and $\mathbf{L}_j(t) \preceq \boldsymbol{\mu}_j$, hence $\boldsymbol{\mu}_i \preceq \boldsymbol{\mu}_j$ and $V_i \subseteq V_j$, where $V_a = \prod_{c=1}^d [r_c, \mu_{a,c}]$ denotes the box of arm a. Let $V = \bigcup_{a \in G_{\ell-1}^\star} V_a$. If $j \notin G_{\ell-1}^\star$, then $\Delta(j \mid G_{\ell-1}^\star) = \lambda_d(V_j \setminus V) \geq \lambda_d(V_i \setminus V) = \Delta(i_\ell^\star \mid G_{\ell-1}^\star)$, so j is also a maximizer at stage ℓ, which contradicts uniqueness. If $j \in G_{\ell-1}^\star$, then $V_i \subseteq V_j \subseteq V$ and $\Delta(i_\ell^\star \mid G_{\ell-1}^\star) = 0$; since marginal gains are non-negative, every arm outside $G_{\ell-1}^\star$ is then a maximizer at stage ℓ, and there are at least $n - \ell + 1 \geq 2$ such arms, which again contradicts uniqueness. □

Part (i) completes the justification of Corollary A.1: the greedy step of Algorithm 1 runs on A_t, so Fact A.1, applied on the ground set A_t, gives $\mathrm{HV}_t^{\mathrm{UCB}}(S_t) \geq \alpha \max_{S \subseteq A_t,\, |S|=k} \mathrm{HV}_t^{\mathrm{UCB}}(S)$, and by (i) this maximum is the one over all size-k subsets of $[n]$. Part (ii) guarantees that, on $\mathcal{E}$, the arms of $G^\star$ always belong to the candidate set of the greedy step, which is used in Appendix A.7.

A.4. Optimism and reduction to optimism error

Recall $u_{i,j}(t) = \min\{1, \widehat{\mu}_{i,j}(t-1) + \beta_i(t)\}$ and

$$\mathrm{HV}_t^{\mathrm{UCB}}(S) = \lambda_d\Big(\bigcup_{i\in S}[r_1, u_{i,1}(t)] \times \cdots \times [r_d, u_{i,d}(t)]\Big).$$

On $\mathcal{E}$, $\mu_{i,j} \le u_{i,j}(t)$ for all i, j (clipping preserves this inequality since $\mu_{i,j} \le 1$), so by monotonicity $\mathrm{HV}(S) \le \mathrm{HV}_t^{\mathrm{UCB}}(S)$ for every S. Let $S^\star \in \arg\max_{|S|=k} \mathrm{HV}(S)$, with $V^\star = \mathrm{HV}(S^\star)$ — this is the benchmark appearing in the regret definition (2), and the only role $S^\star$ plays in this appendix. Since S_t^{opt} maximizes $\mathrm{HV}_t^{\mathrm{UCB}}$ over *all* size-k sets, it dominates $S^\star$ under this score, so by Corollary A.1,

$$\mathrm{HV}_t^{\mathrm{UCB}}(S_t) \;\ge\; \alpha\,\mathrm{HV}_t^{\mathrm{UCB}}(S_t^{\mathrm{opt}}) \;\ge\; \alpha\,\mathrm{HV}_t^{\mathrm{UCB}}(S^\star).$$

Therefore, on $\mathcal{E}$,

$$\bar{r}_t = \alpha\,V^\star - \mathrm{HV}(S_t) \;\le\; \alpha\,\mathrm{HV}_t^{\mathrm{UCB}}(S^\star) - \mathrm{HV}(S_t) \;\le\; \mathrm{HV}_t^{\mathrm{UCB}}(S_t) - \mathrm{HV}(S_t). \tag{A.1}$$

Summing over $t \in [T]$,

$$\bar{R}_T \;\le\; \sum_{t=1}^{T}\big(\mathrm{HV}_t^{\mathrm{UCB}}(S_t) - \mathrm{HV}(S_t)\big) \qquad \text{on } \mathcal{E}. \tag{A.2}$$

A.5. Lipschitz control of the optimism error

Lemma A.4 (Lipschitz bound for hypervolume under coordinate-wise shifts)**.** *There is a constant C_d (one may take $C_d \le 2d$ when rewards lie in $[0,1]^d$, $\mathbf{r} = \mathbf{0}$ and the optimistic vectors are clipped to $[0,1]^d$) such that, on $\mathcal{E}$, for every round t and **every finite set** $S \subseteq [n]$ (not necessarily S_t),*

$$\mathrm{HV}_t^{\mathrm{UCB}}(S) - \mathrm{HV}(S) \;\le\; C_d \sum_{i\in S}\beta_i(t).$$

Proof. Write $S = \{i_1, \dots, i_m\}$. Define the intermediate hypervolume A_ℓ where arms $i_1, \dots, i_\ell$ use optimistic vectors $\mathbf{u}_{i_p}(t)$ and arms $i_{\ell+1}, \dots, i_m$ use true means $\boldsymbol{\mu}_{i_p}$, so $\mathrm{HV}_t^{\mathrm{UCB}}(S) - \mathrm{HV}(S) = \sum_{\ell=1}^{m}(A_\ell - A_{\ell-1})$. Each term changes only arm i_ℓ from $\boldsymbol{\mu}_{i_\ell}$ to $\mathbf{u}_{i_\ell}(t) = \boldsymbol{\mu}_{i_\ell} + \boldsymbol{\delta}_{i_\ell}$ with $0 \le \delta_{i_\ell,j} \le 2\beta_{i_\ell}(t)$: indeed, on $\mathcal{E}$, $\widehat{\mu}_{i,j}(t-1) - \mu_{i,j} \in [-\beta_i(t), \beta_i(t)]$, so $u_{i,j}(t) - \mu_{i,j} \in [0, 2\beta_i(t)]$, and clipping to $[0,1]$ can only reduce this difference while keeping it non-negative. Since all vectors involved lie in $[0,1]^d$, the added hypervolume is contained in a union of d axis-aligned slabs of width at most $2\beta_{i_\ell}(t)$ in one coordinate and at most 1 in the others, so $A_\ell - A_{\ell-1} \le 2d\,\beta_{i_\ell}(t)$. Summing over ℓ gives the claim with $C_d = 2d$. Crucially, this argument never uses $|S| = k$ nor any property of how S was selected, hence it holds for an arbitrary finite S. □

Combining (A.2) with Lemma A.4 applied to $S = S_t$,

$$\bar{R}_T \;\le\; C_d \sum_{t=1}^{T}\sum_{i\in S_t}\beta_i(t) \qquad \text{on } \mathcal{E}. \tag{A.3}$$

The rest of the proof bounds the right-hand side of (A.3) in two ways.

A.6. A gap-free bound

Let $N_i(T) = \sum_{t=1}^{T}\mathbf{1}\{i \in S_t\}$. Reordering (A.3) by arm,

$$\sum_{t=1}^{T}\sum_{i\in S_t}\beta_i(t) = \sum_{i=1}^{n}\sum_{s=1}^{N_i(T)}\sqrt{\frac{2\eta\log(ndT^2/\delta)}{s}} \;\le\; 2\sqrt{2\eta\log(ndT^2/\delta)}\sum_{i=1}^{n}\sqrt{N_i(T)},$$

using $\sum_{s=1}^{m} s^{-1/2} \le 2\sqrt{m}$. Since exactly k arms are pulled per round, $\sum_i N_i(T) = kT$, and Cauchy–Schwarz gives $\sum_i \sqrt{N_i(T)} \le \sqrt{nkT}$. Hence, on $\mathcal{E}$, $\bar{R}_T \le O\big(C_d\sqrt{nkT\log T}\big)$.

Theorem 1 (Gap-free bound (detailed version)). *On $\mathcal{E}$, $\bar{R}_T = O\big(C_d \sqrt{nkT \log T}\big)$.*

Remark. *This argument never used that S_t is built by comparing marginal gains $\Delta_t^{\mathrm{UCB}}(i \mid S)$ across candidates: it only used $|S_t| = k$ and the parametric decay $\beta_i(t) = \Theta(1/\sqrt{N_i(t-1)})$. It is "gap-free" precisely because it is blind to how close arms are, hence to the selection mechanism itself.*

A.7. A gap-dependent bound

The key idea: the algorithm's own greedy construction at round t progressively learns a stage-by-stage benchmark sequence. At stage ℓ, it must identify the best arm conditionally on the previously identified greedy prefix.

A.7.1. The greedy optimal set

Let $G^\star = \{i_1^\star, \dots, i_k^\star\}$ be a greedy construction of an optimal size-k set for the *true* hypervolume:

$$i_\ell^\star \in \arg\max_{i \notin G_{\ell-1}^\star} \Delta(i \mid G_{\ell-1}^\star), \qquad G_\ell^\star = G_{\ell-1}^\star \cup \{i_\ell^\star\}, \quad G_0^\star = \emptyset,$$

with $\Delta(i \mid S) = \mathrm{HV}(S \cup \{i\}) - \mathrm{HV}(S)$.

Assume the greedy maximizer is unique at every stage of $G^\star$'s construction, and define the stage-ℓ gaps relative to this benchmark:

$$\Delta_\ell(i) = \Delta(i_\ell^\star \mid G_{\ell-1}^\star) - \Delta(i \mid G_{\ell-1}^\star) > 0, \quad i \neq i_\ell^\star, \qquad \Delta_{\min} = \min_{\ell \in [k]} \min_{i \neq i_\ell^\star} \Delta_\ell(i).$$

Remark. *By submodularity (Lemma A.1), the sequence of greedy* values *$v_\ell := \Delta(i_\ell^\star \mid G_{\ell-1}^\star)$ is non-increasing: for $i \notin G_\ell^\star$, $G_{\ell-1}^\star \subseteq G_\ell^\star$ gives $\Delta(i \mid G_\ell^\star) \le \Delta(i \mid G_{\ell-1}^\star)$, so $v_{\ell+1} \le v_\ell$. This only constrains the* best *marginal gain at each stage, not the runner-up: it does* not *imply that the gap sequence $\gamma_\ell := \min_{i \neq i_\ell^\star} \Delta_\ell(i)$ is monotonic in ℓ. We therefore use the safe, uniform bound $\Delta_{\min}$ rather than a stage-wise one.*

A.7.2. Deviation of optimistic marginal gains

For $S \subseteq [n]$ and $i \notin S$, write $\Delta_t^{\mathrm{UCB}}(i \mid S) = \mathrm{HV}_t^{\mathrm{UCB}}(S \cup \{i\}) - \mathrm{HV}_t^{\mathrm{UCB}}(S)$.

Lemma A.5 (Deviation of optimistic marginal gains). *On $\mathcal{E}$, for every $S \subseteq [n]$ and $i \notin S$,*

$$\Delta_t^{\mathrm{UCB}}(i \mid S) \le \Delta(i \mid S) + C_d \sum_{j \in S \cup \{i\}} \beta_j(t), \qquad \Delta_t^{\mathrm{UCB}}(i \mid S) \ge \Delta(i \mid S) - C_d \sum_{j \in S} \beta_j(t).$$

Proof. By Lemma A.4 applied to $S \cup \{i\}$ and to S, $\mathrm{HV}_t^{\mathrm{UCB}}(S \cup \{i\}) \le \mathrm{HV}(S \cup \{i\}) + C_d \sum_{j \in S \cup \{i\}} \beta_j(t)$, while optimism gives $\mathrm{HV}_t^{\mathrm{UCB}}(S) \ge \mathrm{HV}(S)$; subtracting gives the upper bound. Symmetrically, $\mathrm{HV}_t^{\mathrm{UCB}}(S \cup \{i\}) \ge \mathrm{HV}(S \cup \{i\})$ and $\mathrm{HV}_t^{\mathrm{UCB}}(S) \le \mathrm{HV}(S) + C_d \sum_{j \in S} \beta_j(t)$ give the lower bound. □

A.7.3. Per-round deviation stage

Write the algorithm's own greedy construction at round t as a chain $\hat{S}_{t,0} = \emptyset \subset \hat{S}_{t,1} \subset \cdots \subset \hat{S}_{t,k} = S_t$, where $\hat{S}_{t,\ell} = \hat{S}_{t,\ell-1} \cup \{\hat{\imath}_{t,\ell}\}$ and $\hat{\imath}_{t,\ell} \in \arg\max_{i \in A_t \setminus \hat{S}_{t,\ell-1}} \Delta_t^{\mathrm{UCB}}(i \mid \hat{S}_{t,\ell-1})$.[1] Define the *first deviation stage*, comparing the algorithm's trajectory to the benchmark $G^\star$:

$$\ell^\dagger(t) = \min\{\ell \in [k] : \hat{S}_{t,\ell} \neq G_\ell^\star\},$$

with $\ell^\dagger(t) = +\infty$ if $\hat{S}_{t,\ell} = G_\ell^\star$ for every $\ell \in [k]$ (i.e. $S_t = G^\star$ exactly). This partitions $[T]$ into

$$\mathrm{Match} = \{t : \ell^\dagger(t) = \infty\}, \qquad \mathrm{Dev}_\ell = \{t : \ell^\dagger(t) = \ell\}\ (\ell \in [k]), \qquad [T] = \mathrm{Match} \sqcup \bigsqcup_{\ell=1}^{k} \mathrm{Dev}_\ell.$$

[1] By Lemma A.3(ii), on $\mathcal{E}$ and under the uniqueness assumption, $G^\star \subseteq A_t$: the safe-pruning step never removes an arm of the greedy benchmark, so $i_\ell^\star$ is always a candidate at stage ℓ.

By construction, $t \in \mathrm{Dev}_\ell$ means $\hat{S}_{t,\ell-1} = G^\star_{\ell-1}$ — this is built into the partition, no induction over earlier rounds is needed.

Since $\bar{R}_T = \sum_{t=1}^T \bar{r}_t$ exactly by (2), this partition refines the same per-round sum bounded pointwise in (A.1): rather than controlling every term $\bar{r}_t$ uniformly via Lemma A.4 and a single global Cauchy–Schwarz step (as in Appendix A.6), we now split

$$\bar{R}_T \;=\; \sum_{t\in\mathrm{Match}} \bar{r}_t \;+\; \sum_{\ell=1}^{k} \sum_{t\in\mathrm{Dev}_\ell} \bar{r}_t, \tag{A.4}$$

and bound the two kinds of terms separately: we show below that the first sum is non-positive; the remainder of this subsection bounds the second.

A.7.4. Regret on matched rounds is non-positive

We bound the first term of (A.4). If $t \in \mathrm{Match}$, then $S_t = G^\star$. By Fact A.1 with $f = \mathrm{HV}$, $G = G^\star$,

$$\bar{r}_t = \alpha\, V^\star - \mathrm{HV}(S_t) = \alpha\, V^\star - \mathrm{HV}(G^\star) \;\le\; \alpha\, V^\star - \alpha\, V^\star = 0.$$

A.7.5. Witness-counting bound on deviation rounds

Fix ℓ and $t \in \mathrm{Dev}_\ell$, and write $i := \hat{\imath}_{t,\ell} \neq i^\star_\ell$. Since i was chosen over $i^\star_\ell$ at stage ℓ of round t, $\Delta^{\mathrm{UCB}}_t(i \mid G^\star_{\ell-1}) \ge \Delta^{\mathrm{UCB}}_t(i^\star_\ell \mid G^\star_{\ell-1})$. Applying Lemma A.5 at $S = G^\star_{\ell-1}$,

$$\Delta(i \mid G^\star_{\ell-1}) + C_d \sum_{j\in G^\star_{\ell-1}\cup\{i\}} \beta_j(t) \;\ge\; \Delta(i^\star_\ell \mid G^\star_{\ell-1}) - C_d \sum_{j\in G^\star_{\ell-1}} \beta_j(t),$$

so, using $|G^\star_{\ell-1}| = \ell - 1$,

$$\Delta_\ell(i) \;\le\; C_d\Big(2 \sum_{j\in G^\star_{\ell-1}} \beta_j(t) + \beta_i(t)\Big) \;\le\; C_d(2\ell-1) \max_{j\in G^\star_{\ell-1}\cup\{i\}} \beta_j(t).$$

Hence there exists a *witness* $w(t) \in G^\star_{\ell-1} \cup \{i\}$ (at most ℓ candidates) with

$$\beta_{w(t)}(t) \;\ge\; \frac{\Delta_\ell(i)}{C_d(2\ell-1)} \;\ge\; \frac{\Delta_{\min}}{C_d(2\ell-1)},$$

$$\text{so} \qquad N_{w(t)}(t-1) \;\le\; m_\ell := \frac{2\eta C_d^2 (2\ell-1)^2 \log(ndT^2/\delta)}{\Delta^2_{\min}} = O\!\left(\frac{C_d^2\, \ell^2 \log T}{\Delta^2_{\min}}\right).$$

Since $G^\star_{\ell-1} \cup \{i\} = \hat{S}_{t,\ell} \subseteq S_t$, the witness $w(t)$ is pulled at round t. Group Dev_ℓ by witness identity: for fixed $w \in [n]$, every $t \in \mathrm{Dev}_\ell$ with $w(t) = w$ increments N_w by exactly 1 while $N_w(t-1) \le m_\ell$; there are at most $m_\ell + 1$ such rounds. Summing over $w \in [n]$,

$$|\mathrm{Dev}_\ell| \;\le\; n\,(m_\ell + 1) \;=\; O\!\left(\frac{C_d^2\, n\ell^2 \log T}{\Delta^2_{\min}}\right). \tag{A.5}$$

A.7.6. Bounding total regret on deviation rounds

We now bound the second term of (A.4), $\sum_{\ell=1}^k \sum_{t\in\mathrm{Dev}_\ell} \bar{r}_t$. A naive approach would bound $\bar{r}_t \le 1$ on every round of Dev_ℓ and multiply by the count $|\mathrm{Dev}_\ell| = O(C_d^2\, n\ell^2 \log T/\Delta^2_{\min})$ from (A.5), giving a regret scaling as $1/\Delta^2_{\min}$. We avoid this by bounding the *total* regret over Dev_ℓ directly, via the same Cauchy–Schwarz idea as in Appendix A.6 — applied *locally* to Dev_ℓ rather than to the full horizon. Since that argument scales as the *square root* of the number of rounds involved, it converts the $1/\Delta^2_{\min}$ scale of the count into a $1/\Delta_{\min}$ contribution to regret.

On $\mathcal{E}$, the pointwise bound (A.1) together with Lemma A.4 gives, for every round t,

$$\bar{r}_t \;\le\; \mathrm{HV}^{\mathrm{UCB}}_t(S_t) - \mathrm{HV}(S_t) \;\le\; C_d \sum_{i\in S_t} \beta_i(t).$$

Fix $\ell \in [k]$, and for $i \in [n]$ let $D_i = \{t \in \mathrm{Dev}_\ell : i \in S_t\}$ be the (possibly empty) set of rounds, among those at which the trajectory first departs from $G^\star$ at stage ℓ, where arm i happens to be pulled. The pull count $N_i(t-1)$ increases by exactly 1 at each pull of i, so as t ranges over D_i (a set of pulls of i), the values $N_i(t-1)$ are pairwise distinct; hence their j-th smallest value is at least $j-1$. Since $\beta_i(t) \propto N_i(t-1)^{-1/2}$ is decreasing, this bounds the sum by the worst case where D_i consists of arm i's *earliest* $|D_i|$ pulls:

$$\sum_{t\in D_i} \beta_i(t) \;\le\; \sum_{s=1}^{|D_i|} \sqrt{\frac{2\eta \log(ndT^2/\delta)}{s}} \;\le\; 2\sqrt{2\eta \log(ndT^2/\delta)}\,\sqrt{|D_i|} \;=\; O\big(\sqrt{|D_i| \log T}\big),$$

using $\sum_{s=1}^{m} s^{-1/2} \le 2\sqrt{m}$ and $\log(ndT^2/\delta) = O(\log T)$ — the same argument as in Appendix A.6, restricted to $D_i \subseteq \{\text{pulls of } i\}$ instead of all of i's pulls over $[T]$.

Hence

$$\sum_{t\in \mathrm{Dev}_\ell} \sum_{i\in S_t} \beta_i(t) = \sum_{i=1}^{n} \sum_{t\in D_i} \beta_i(t) = O\Big(\sqrt{\log T} \sum_{i=1}^{n} \sqrt{|D_i|}\Big).$$

Since $|S_t| = k$ for every t, we have $\sum_{i=1}^{n} |D_i| = \sum_{t\in \mathrm{Dev}_\ell} \sum_{i=1}^{n} \mathbf{1}\{i \in S_t\} = \sum_{t\in \mathrm{Dev}_\ell} |S_t| = k|\mathrm{Dev}_\ell|$. Next, by Cauchy–Schwarz (with $a_i = 1$, $b_i = \sqrt{|D_i|}$),

$$\sum_{i=1}^{n} \sqrt{|D_i|} \;\le\; \sqrt{n}\,\sqrt{\sum_{i=1}^{n} |D_i|} \;=\; \sqrt{nk|\mathrm{Dev}_\ell|}.$$

Therefore

$$\sum_{t\in \mathrm{Dev}_\ell} \bar{r}_t \;\le\; C_d \sum_{t\in \mathrm{Dev}_\ell} \sum_{i\in S_t} \beta_i(t) \;=\; O\big(C_d \sqrt{nk|\mathrm{Dev}_\ell| \log T}\big) \;=\; O\left(\frac{C_d^2\, n\ell \sqrt{k}\, \log T}{\Delta_{\min}}\right),$$

using (A.5) in the last step. Summing over $\ell \in [k]$ (using $\sum_{\ell=1}^{k} \ell = O(k^2)$) and combining with (A.4) and the non-positive first term,

$$\bar{R}_T \;\le\; \sum_{\ell=1}^{k} O\left(\frac{C_d^2\, n\ell \sqrt{k} \log T}{\Delta_{\min}}\right) \;=\; O\left(\frac{C_d^2\, nk^{2.5} \log T}{\Delta_{\min}}\right) \qquad \text{on } \mathcal{E}.$$

Theorem 2 (Gap-dependent bound (detailed version))**.** *On $\mathcal{E}$, $\bar{R}_T = O\big(C_d^2\, nk^{2.5} \log T/\Delta_{\min}\big)$.*

A.8. Combining both bounds

Since rewards lie in $[0,1]^d$, $\bar{r}_t \le 1$ and hence $\bar{R}_T \le T$ always (on or off $\mathcal{E}$). Therefore

$$\mathbb{E}[\bar{R}_T] = \mathbb{E}[\bar{R}_T \mathbf{1}_{\mathcal{E}}] + \mathbb{E}[\bar{R}_T \mathbf{1}_{\mathcal{E}^c}] \le \min\Big\{ O\big(C_d \sqrt{nkT \log T}\big),\; O\big(C_d^2\, nk^{2.5} \log T/\Delta_{\min}\big) \Big\} + T\,\mathbb{P}(\mathcal{E}^c).$$

Choosing $\delta = 1/T$ gives $T\,\mathbb{P}(\mathcal{E}^c) \le 1$, so:

Theorem 3 (Regret of *THV-UCB* (detailed version))**.** *With $\delta = 1/T$,*

$$\mathbb{E}[\bar{R}_T] \;\le\; \min\Big\{ O\big(C_d \sqrt{nkT \log T}\big),\; O\big(C_d^2\, nk^{2.5} \log T/\Delta_{\min}\big) \Big\} + O(1).$$

In particular $\limsup_{T\to\infty} \mathbb{E}[\bar{R}_T]/T \le 0$*: the α-approximation regret of THV-UCB is sublinear on every instance.*

A.9. Discussion: what does each bound use from the algorithm?

The two guarantees of Theorem 3 are not in competition; they cover complementary regimes, exactly as the minimax and gap-dependent bounds do for classical K-armed UCB. The gap-free bound (Theorem 1) is *algorithm-agnostic*: its proof only uses that k arms are pulled per round and that confidence widths shrink at rate $1/\sqrt{N_i(t-1)}$; it never degrades, even as $\Delta_{\min} \to 0$, the regime where the gap-dependent bound diverges. Conversely, the gap-dependent bound (Theorem 2) is *mechanism-aware*: it tracks, round by round, the exact comparisons $\Delta_t^{\mathrm{UCB}}(\cdot \mid G^\star_{\ell-1})$ that the greedy step of *THV-UCB* performs against the benchmark trajectory $G^\star$ — note that $G^\star$, unlike $S^\star$, never appears in the regret's definition; it is purely an internal proof device for tracking the algorithm's stage-wise progress. Its proof combines two ingredients: a witness-counting argument bounding how many rounds can deviate from $G^\star$ at each stage (see (A.5)), and a Cauchy–Schwarz argument bounding the *total* regret accrued on those rounds.

References

[1] G. L. Nemhauser, L. A. Wolsey, M. L. Fisher, An analysis of approximations for maximizing submodular set functions—I, Mathematical Programming 14 (1) (1978) 265–294.